

ZECHENG WANG

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Education

- **Boston University**, PhD in Computer Science, 2024 - present
 - Area: Machine Learning, World Models, Autonomous Systems
- **New York University Shanghai**, B.S. in Computer Science; B.S. in Data Science, 2023
 - Major GPA: 3.93; Cumulative GPA: 3.90; *magna cum laude*

Research Experience

- **Graduate Research Assistant**, Boston University, Boston, MA, Sept 2024 - present
 - Develop and analyze an energy-based world model to capture multi-mode futures via iterative latent inference
 - Developed a higher-fidelity BeamNG benchmark for *CrashCourse* to test agile evasion, stabilization, and recovery
 - Developed a general DevCV data pipeline and constructed the *BabyVLM-V2* Picture Vocabulary dataset
- **Research Intern**, part-time, Microsoft Research, Redmond, Oct 2024 - June 2025
 - Studied in-context learning (ICL) in vision-language-action models (VLAs) for ID/OOD robotics-task generalization
 - Implemented a framework for incorporating in-context demonstrations during training and execution
 - Analyzed the key components needed to enable ICL for ID/OOD generalization in robotics

Mentor: Zhengyuan Yang, Microsoft Research.

- **Research Assistant**, full-time, NYU Abu Dhabi, Aug 2023 - Aug 2024
 - Reviewed pre-training, cross-environment learning, and scalable reinforcement-learning architectures
 - Designed algorithms and experiments and implemented code to evaluate research ideas
 - Wrote manuscripts for publication and technical reports for communication with the group and collaborators

Advisor: Prof. Keith Ross, NYU Abu Dhabi.

Publications

- S. Wang*, W. Wang*, **Z. Wang***, M. Whittan*, et al., “BabyVLM-V2: Toward Developmentally Grounded Pretraining and Benchmarking of Vision Foundation Models,” *IEEE/CVF CVPR 2026*, Denver, CO, USA, pp. 23696–23708.
- **Z. Wang***, C. Wang*, Z. Dong*, and K. W. Ross, “Pre-training with Synthetic Data Helps Offline Reinforcement Learning,” *ICLR 2024*, Vienna, Austria.
- **Z. Wang** and Y.-C. Tam, “Suffix Retrieval-Augmented Language Modeling,” *IEEE ICASSP 2023*, Rhodes Island, Greece, pp. 1–5, doi: 10.1109/ICASSP49357.2023.10096450.
- Y.-C. Tam, J. Xu, J. Zou, **Z. Wang**, T. Liao, and S. Yuan, “Robust Unstructured Knowledge Access in Conversational Dialogue with ASR Errors,” *IEEE ICASSP 2022*, Singapore, Singapore, pp. 6702–6706, doi: 10.1109/ICASSP43922.2022.9746741.

Preprints & Manuscripts under Review

- M. Whittan, **Z. Wang***, P. Liu*, Q. T. Truong, S. Wang, et al., “Making Sense of Touch from the Child’s View for Contrastive Learning,” manuscript accepted at *2026 IEEE International Conference on Development and Learning (ICDL 2026)*. *arXiv:2606.31943*, 2026.
- R. Sajjadinasab, Z. Wei, **Z. Wang**, D. Bordal, J. Wu, M. Wei, L. Lai, and E. Ohn-Bar, “CrashCourse: Towards Agility in Safety-Critical Driving,” manuscript under review at the *10th Conference on Robot Learning (CoRL 2026)*, 2026.

Teaching Experience

- **Teaching Fellow**, part-time, Boston University, Fall 2025; Spring 2026; Fall 2026
 - Assisted with CS 542: Principles of Machine Learning in Fall 2025 and Spring 2026; appointed for Fall 2026
 - Led weekly labs and office hours on problem sets, Bayesian statistics, backpropagation, and other core concepts
 - Managed course logistics, including assignment distribution, exam coordination, and grading support
- **Teaching Fellow**, part-time, Boston University RISE Pre-College Program, Summers 2025 and 2026
 - **2026**: Teach an ML practicum with the instructor, giving lectures, designing assignments, and answering questions
 - **2025**: Supervised high school research projects contributing to *BabyVLM-V2* (CVPR 2026)
 - **2025**: Mentored students using vision-language and object-detection APIs to build data-mining pipelines
 - **2025**: Guided projects from ideation and implementation to poster presentation at the RISE Symposium
- **Teaching Assistant**, part-time, NYU Abu Dhabi, Oct 2023 - Aug 2024
 - Assisted with the Machine Learning course
 - Designed and graded theoretical and coding assignments and exam questions
 - Delivered PyTorch tutorials and answered student questions

Presentations

- **Poster**, *New England Computer Vision Workshop (NECV)*, UMass Amherst, MA, Nov 2025.
- **Poster**, *The Twelfth International Conference on Learning Representations (ICLR 2024)*, Vienna, Austria: "Pre-training with Synthetic Data Helps Offline Reinforcement Learning."
- **Oral presentation**, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2023)*, Rhodes, Greece: "Suffix Retrieval-Augmented Language Modeling."
- **Poster**, *Tandon Research Excellence Exhibit*, NYU Tandon School of Engineering, New York, Spring 2022: "AI for Scientific Research: Bio-treatment."
- **Poster**, *NYU Shanghai Undergraduate Research Symposium*, Shanghai, Fall 2021: "Knowledge-grounded Task-oriented Dialog Modeling on Spoken Conversations."

Selected Research Projects

- *Energy-Based World Models for Multimodal Future Prediction*, Boston University, 2026 - present
 - Formulated and led an EBM study of multimodal driving futures against a deterministic DINO-WM baseline
 - Implemented training and iterative latent refinement from noise, prior predictions, and real-scene initializations
 - Separated mode representation, reachability, and context selection, exposing persistence and donor-content failures
- *CrashCourse: Towards Agility in Safety-Critical Driving - BeamNG Benchmark*, Boston University, 2026
 - Developed the BeamNG benchmark to test driving policies under higher-fidelity dynamics and safety-critical events
 - Implemented high-speed crash and near-crash scenarios requiring evasion, stabilization, and recovery
 - Built scenario execution, damage scoring, and policy evaluation; produced results for the supplementary material
- *BabyVLM-V2: Picture Vocabulary Dataset*, Boston University, 2025
 - Constructed the Picture Vocabulary dataset by adapting the NIH Baby Toolbox to naturalistic egocentric data
 - Built an object-detection pipeline for data mining, filtering, cleaning, and quality control
 - Generalized the pipeline to additional benchmark tasks and wrote the corresponding paper section
- *Cross-Environment Learning with Text Description and Scalable Transformer Architecture*
 - Studied cross-environment knowledge transfer using text descriptions of states and actions
 - Evaluated the Q-Transformer on high-dimensional action spaces for knowledge transfer
 - Proposed a novel Transformer architecture for scalable actor-critic updates

Advisor: Prof. Keith Ross, NYU Abu Dhabi

- *Pre-training with Synthetic Data Helps Offline Reinforcement Learning*, NYU Abu Dhabi, 2023
 - Proposed a simple, effective synthetic pre-training scheme for Decision Transformer and MLP architectures
 - Outperformed prior language pre-training on D4RL MuJoCo using 3% pre-training and 67% fine-tuning compute
 - Published at ICLR 2024

Advisor: Prof. Keith Ross, NYU Abu Dhabi

- *Suffix Retrieval-Augmented Language Modeling*, NYU Shanghai, 2022-2023
 - Proposed prefix-suffix retrieval from external knowledge to simulate bidirectionality in causal language modeling
 - Reduced perplexity by 20% on DSTC9 compared with popular pre-trained language-model baselines
 - Published at ICASSP 2023

Advisor: Prof. Yik-Cheung Tam, NYU Shanghai

- *AI for Scientific Research: Bio Treatment*, 2022
 - Framed vaccine-efficacy prediction as a machine-learning problem
 - Implemented a flexible ML pipeline that improved performance
 - Enhanced model interpretability on local and global scales

Mentor: Dr. Sergey Samsonau, NYU Tandon School of Engineering

Honors and Awards

- Dean's List for Academic Year 2019, 2020, 2021, 2022
- Dean's Undergraduate Research Funding 2021, 2022
- Major honors in Computer Science

Leadership Experience

- **ML Lead & Advisor**, AI for Scientific Research, 2022-present
Mentor: Dr. Sergey Samsonau, NYU Tandon School of Engineering

Relevant Skills

- Programming Languages: proficient in Python; basic knowledge of Java, C, and R
- Machine Learning: PyTorch, scikit-learn, OpenCV, Elasticsearch
- Systems: Linux, SLURM
- Languages: proficient in English; fluent in Chinese